

Lecture Number One

May I welcome you all to this series of lectures. The large enrollment is quite a compliment to the Institute, and perhaps to the lecturer; but it also poses something of a problem. We shall not be able to handle this course on an informal or round-table basis. However, I should like to welcome as much discussion and as many intelligent questions as we can get, but I shall have to reserve the right to cut short discussion or not to answer questions in the interest of getting along with the course. You all understand our problem, I am sure.

I hope you will find that your time and money will be profitably spent in this course; but I want to add that the purpose of this course is to provide illustrative examples and discussions only, and not to supply practical ideas for security market operations. We assume no responsibility for anything said along the latter lines in this course; and so far as our own business is concerned we may or we may not have an interest in any of the securities that are mentioned and discussed. That is also a teaching problem with which we have been familiar through the years, and we want to get it behind us as soon as we can.

The subject of this course is "Current Problems in Security Analysis", and that covers a pretty wide field. Actually, the idea is to attempt to bring our textbook "Security Analysis" up to date, in the light of the experience of the last six years since the 1940 revision was published.

The subject matter of security analysis can be divided in various ways. One division might be in three parts: First, the techniques of security analysis; secondly, standards of safety and common stock valuation; and thirdly, the relationship of the analyst to the security market.

Another way of dividing the subject might be to consider, first, the analyst as an investigator, in which role he gathers together all the relevant facts and serves them up in the most palatable and illuminating fashion he can. And then to consider the analyst as a judge of values, or an evaluator. This first division of the subject is rather useful, I think, because there is a good field in Wall Street for people whose work it will be mainly to digest the facts, and to abstain from passing judgment on the facts, leaving that to other people.

Such sticking to the facts alone might be very salutary; for the judgment of security analysts on securities is so much influenced by market conditions down here that most of us are not able, I fear, to express valuation judgments as good analysts. We find ourselves almost always acting as a mixture of market experts and security experts. I had hoped that there would be some improvement in that situation over the years, but I must confess that I haven't seen a great deal of it. Analysts have recently been acting in Wall Street pretty much as they always have, that is to say, with one eye on the balance sheet and income account, and the other eye on the stock ticker. It might be best in this introductory lecture to deal with the third aspect of the security analyst's work, and that is his relationship to the security market. It is a little more interesting, perhaps, than the other subdivisions, and I think it is relevant as introductory material.

The correct attitude of the security analyst toward the stock market might well be that of a man toward his wife. He shouldn't pay too much attention to what the lady says, but he can't afford to ignore it entirely. That is pretty much the position that most of us find ourselves vis-à-vis the stock market.

When we consider how the stock market has acted in the last six years, we shall conclude that it has acted pretty much as one would expect it to, based upon past experience. To begin with, it has gone up and it has gone down, and different securities have acted in different fashion. We have tried to illustrate this simply, by indicating on the blackboard the behavior of some sample stocks since the end of 1938. Let me take occasion

to point out some of the features in this record that may interest security analysts.

There are two elements of basic importance, I think, that the analyst should recognize in the behavior of stocks over the last six years. The first is the principle of continuity, and the other is what I would call the principle of deceptive selectivity in the stock market.

First, with regard to continuity: The extraordinary thing about the securities market, if you judge it over a long period of years, is the fact that it does not go off on tangents permanently, but it remains in continuous orbit. When I say that it doesn't go off on tangents, I mean the simple point that after the stock market goes up a great deal it not only comes down a great deal but it comes down to levels to which we had previously been accustomed. Thus we have never found the stock market as a whole going off into new areas and staying there permanently because there has been a permanent change in the basic conditions. I think you would have expected such new departures in stock prices. For the last thirty years, the period of time that I have watched the securities market, we have had two world wars; we have had a tremendous boom and a tremendous deflation; we now have the Atomic Age on us. Thus you might well assume that the security market could really have been permanently transformed at one time or another, so that the past records might not have been very useful in judging future values.

These remarks are relevant, of course, to developments since 1940. When the security market advanced in the last few years to levels which were not unexampled but which were high in relation to past experience, there was a general tendency for security analysts to assume that a new level of values had been established for stock prices which was quite different from those we had previously been accustomed to. It may very well be that individual stocks as a whole are worth more than they used to be. But the thing that doesn't seem to be true is that they are worth so much more than they used to be that past experience -- i.e., past levels and patterns of behavior -- can be discarded.

One way of expressing the principle of continuity in concrete terms would be as follows: When you look at the stock market as a whole, you will find from experience that after it has advanced a good deal it not only goes down -- that is obvious -- but it goes down to levels substantially below earlier high levels. Hence it has always been possible to buy stocks at lower prices than the highest of previous moves, not of the current move. That means, in short, that the investor who says he does not wish to buy securities at high levels, because they don't appeal to him on a historical basis or on an analytical basis, can point to past experience to warrant the assumption that he will have an opportunity to buy them at lower prices -- not only lower than current high prices, but lower than previous high levels. In sum, therefore, you can take previous high levels, if you wish, as a measure of the danger point in the stock market for investors, and I think you will find that past experience would bear you out using this as a practical guide. Thus, if you look at this chart of the Dow Jones Industrial Average, you can see there has never been a time in which the price level has broken out, in a once-for-all or permanent way, from its past area of fluctuations. That is the thing I have been trying to point out in the last few minutes.

Another way of illustrating the principle of continuity is by looking at the long-term earnings of the Dow-Jones Industrial Average. We have figures here running back to 1915, which is more than thirty years, and it is extraordinary to see the persistence with which the earnings of the Dow-Jones Industrial Average return to a figure of about \$10 per unit. It is true that they got away from it repeatedly. In 1917, for example, they got up to \$22 a unit; but in 1921 they earned nothing. And a few years later they were back to \$10. In 1915 the earnings of the unit were \$10.59; in 1945 they were practically the same. All of the changes in between appear to have been merely of fluctuations around the central figure. So much for this idea of continuity?

The second thing that I want to talk about is selectivity. Here is an idea that has misled security analysts and advisers to a very great extent. In the few weeks preceding the recent break in the stock market I noticed that a great many of the brokerage house advisers were saying that now that the market has ceased to go up

continuously, the thing to do is to exercise selectivity in your purchases; and in that way you can still derive benefits from security price changes. Well, it stands to reason that if you define selectivity as picking out a stock which is going to go up a good deal later on -- or more than the rest -- you are going to benefit. But that is too obvious a definition. What the commentators mean, as is evident from their actual arguments, is that if you buy the securities which apparently have good earnings prospects, you will then benefit market-wise; whereas if you buy the others you won't.

History shows this to be a very plausible idea but an extremely misleading one; that is why I referred to this concept of selectivity as deceptive. One of the easiest ways to illustrate that is by taking two securities here in the Dow-Jones Average, National Distillers and United Aircraft. You will find that National Distillers sold at lower average prices in 1940-1942 than in 1935-1939. No doubt there was a general feeling that the company's prospects were not good, primarily because it was thought that war would not be a very good thing for a luxury type of business such as whiskey is politely considered to be.

In the same way you will find that the United Aircraft Company through 1940-1942, was better regarded than the average stock, because it was thought that here was a company that had especially good prospects of making money; and so it did. But if you had bought and sold these securities, as most people seem to have done, on the basis of these obvious differential prospects, you would have made a complete error. For, as you see, National Distillers went up from the low of 1940 more than fivefold recently, and is now selling nearly four times its 1940 price. The buyer of United Aircraft would have had a very small profit at its best price and would now have a loss of one third of his money.

This principle of selectivity can be explored in various other ways.

*** Now my point in going at these two things in such detail is to try to bring home to you the fact that what seems to be obvious and simple to the people in Wall Street, as well as to their customers, is not really obvious and simple at all. You are not going to get good results in security analysis by doing the simple, obvious thing of picking out the companies that apparently have good prospects -- whether it be the automobile industry, or the building industry, or any such combination of companies which almost everybody can tell you are going to enjoy good business for a number of years to come. That method is just too simple and too obvious -- and the main fact about it is that it does not work well. The method of selectivity which I believe does work well is one that is based on demonstrated value differentials representing the application of security analysis techniques which have been well established and well tested. These techniques frequently yield indications that a security is undervalued, or at least that it is definitely more attractive than other securities may be, with which it is compared.

As an example of that kind of thing, I might take the comparisons that were made in the Security Analysis*, 1940 edition, between three groups of common stocks. They were compared as of the end of 1938, or just before the war. Of these groups one contained common stocks said to be speculative because their price was high; the second contained those said to be speculative because of their irregular record; and the third contained those said to be attractive investments because they met investment tests from a quantitative standpoint. Let me now mention the names of the stocks, and indicate briefly what is their position as of today. Group A consisted of * "Security Analysis" by Graham & Dodd.

General Electric, Coca-Cola, and Johns-Manville. Their combined price at the end of 1938 was \$281, and at recent lows it was \$203.50 which meant that they have advanced eight per cent. The second group (about which we expressed no real opinion except that they could not be analyzed very well) sold in the aggregate for 124 at the end of 1938 and at recent lows for 150, which was an advance of 20 per cent.

The three stocks which were said to be attractive investments from the quantitative standpoint sold at 70 1/2 at the end of 1938 -- that is for one share of each -- and their value at the recent lows was 207, or an increase

of 190 per cent.

Of course, these performances may be just a coincidence. You can't prove a principle by one or two examples. But I think it is a reasonably good illustration of the results which you should get on the average by using investment tests of merit, as distinct from the emphasis on general prospects which plays so great a part in most of the analysis that I see around the Street.

*** I want to pass on finally to the most vulnerable position of the securities market in the recent rise, and that is the area of new common stock offerings. The aggregate amount of these offerings has not been very large in hundreds of millions of dollars, because the typical company involved was comparatively small. But I think the effect of these offerings upon the position of people in Wall Street was quite significant, because all of these offerings were bought by people who, I am quite sure, didn't know what they were doing and were thus subject to very sudden changes of heart and attitude with regard to their investments. If you made any really careful study of the typical offerings that we have seen in the last twelve months you will agree, I am sure, with a statement made (only in a footnote unfortunately) by the Securities and Exchange Commission on August 20, 1946. They say that: "The rapidity with which many new securities, whose evident hazards are plainly stated in a registration statement and prospectus, are gobbled up at prices far exceeding any reasonable likelihood of return gives ample evidence that the prevalent demand for securities includes a marked element of blind recklessness. Registration cannot cure that."

That is true. Among the astonishing things is the fact that the poorer the security the higher relatively was the price it was sold at. The reason is that most of the sounder securities had already been sold to and held by the public, and their market price was based on ordinary actions of buyers and sellers. The market price of the new securities has been largely determined, I think, by the fact that security salesmen could sell any security at any price; and there was therefore a tendency for the prices to be higher for these new securities than for others of better quality. I think it is worthwhile giving you a little resumé of one of the most recent prospectuses, which is summarized in the Standard Corporation Record of September 13, about a week ago. I don't think this stock was actually sold, but it was intended to be sold at \$16 a share. The name of the company is the Northern Engraving and Manufacturing Company, and we have this simple set-up: There are 250,000 shares to be outstanding, some of which are to be sold at \$16 for the account of stockholders. That meant that this company was to be valued at \$4-million in the market.

Now, what did the new stockholder get for his share of the \$4-million? In the first place, he got \$1,350,000 worth of tangible equity. Hence he was paying three times the amount of money invested in the business. In the second place, he got earnings which can be summarized rather quickly. For the five years 1936-40, they averaged 21 cents a share; for the five years ended 1945, they averaged 65 cents a share. In other words, the stock was being sold at about 25 times the prewar earnings. But naturally there must have been some factor that made such a thing possible, and we find it in the six months ending June 30, 1946, when the company earned \$1.27 a share. In the usual parlance of Wall Street, it could be said that the stock was being sold at six and a half times its earnings, the point being the earnings are at the annual rate of \$2.54, and \$16 is six or seven times that much.

It is bad enough, of course, to offer to the public anything on the basis of a six months' earnings figure alone, when all the other figures make the price appear so extraordinarily high. But in this case it seems to me the situation is extraordinary in another respect -- that it is in relation to the nature of the business. The company manufactures metal nameplates, dials, watch-dials, panels, etc. The products are made only against purchase contracts and are used by manufacturers of motors, controls, and equipment, and so forth.

Now, we don't stress industrial analysis particularly in our course in security analysis, and I am not going to stress it here. But we have to assume that the security analyst has a certain amount of business sense. Surely he would ask himself, "how much profit can a company make in this line of business -- operating on purchase

contracts with automobile and other manufacturers -- in relation both to its invested capital and its sales?"

In the six months ended June 1946 the company earned 15 per cent on its sales after taxes. It had previously tended to earn somewhere around three or four per cent on sales after taxes. It seems to me anyone would know that these earnings for the six months arose from the fact that any product could be sold provided only it could be turned out, and that extremely high profits could be realized in this kind of market. I think it would have been evident that under more sound conditions this is the kind of business which is doomed to earn a small profit margin on its sales and only a moderate amount on its net worth, for it has nothing particular to offer except the know-how to turn out relatively small gadgets for customer buyers.

That, I believe, illustrates quite well what the public had been offered in this recent new security market. There are countless other illustrations that I could give. I would like to mention one that is worth referring to, I think, because of its contrast with other situations.

The Taylorcraft Company is a maker of small airplanes. In June, 1946, they sold 20,000 shares of stock to the public at \$13, the company getting one dollar; and then they voted a four-for-one split up. The stock is now quoted around two and a half or two and three quarters, the equivalent of about \$11 for the stock that was sold.

If you look at the Taylorcraft Company, you find some rather extraordinary things in its picture. To begin with, the company is today selling for about \$3-million, and this is supposedly in a rather weak market. The working capital shown as of June 30, 1946, is only \$103,000. It is able to show even that much working capital, first, after including the proceeds of the sale of this stock, and secondly, after not showing as a current liability an excess profits tax of \$196,000 which they are trying to avoid by means of a "Section 722" claim. Well, practically every corporation that I know of has filed Section 722 claims to try to cut down their excess profits taxes. This is the only corporation I know of that, on the strength of filing that claim, does not show its excess profits tax as a current liability.

They also show advances payable, due over one year, of \$130,000, which of course don't have to be shown as current liabilities. Finally, the company shows \$2,300,000 for stock and surplus, which is not as much as the market price of the stock. But even here we note that the plant was marked up by \$1,150,000, so that just about half of the stock and surplus is represented by what I would call an arbitrary plant mark-up.

Now, there are several other interesting things about the Taylorcraft Company itself, and there are still other things even more interesting when you compare it with other aircraft companies. For one thing, the Taylorcraft Company did not publish reports for a while and it evidently was not in too comfortable a financial position. Thus it arranged to sell these shares of stock in an amount which did not require registration with the SEC. But it is also a most extraordinary thing for a company in bad financial condition to arrange to sell stock to tide it over, and at the same time to arrange to split up its stock four for one. That kind of operation -- to split a stock from \$11 to three dollars -- seems to me to be going pretty far in the direction of trading on the most unintelligent elements in Wall Street stock purchasing that you can find.

But the really astonishing thing is to take Taylorcraft and compare it, let us say, with another company like Curtiss-Wright. Before the split-up, Taylorcraft and Curtiss-Wright apparently were selling about the same price, but that doesn't mean very much. The Curtiss-Wright Company is similar to United Aircraft in that its price is now considerably lower than its 1939 average. The latter was eight and three quarters, and its recent price was five and three quarters. In the meantime, the Curtiss-Wright Company has built up its working capital from a figure perhaps of \$12-million to \$130-million, approximately. It turns out that this company is selling in the market for considerably less than two thirds of its working capital.

The Curtiss-Wright Company happens to be the largest airplane producer in the field, and the Taylorcraft

Company probably is one of the smallest. There are sometimes advantages in small size and disadvantages in large size; but it is hard to believe that a small company in a financially weak position can be worth a great deal more than its tangible investment, when the largest companies in the same field are selling at very large discounts from their working capital. During the period in which Taylorcraft was marking up its fixed assets by means of this appraisal figure, the large companies like United Aircraft and Curtiss-Wright marked down their plants to practically nothing, although the number of square feet which they owned was tremendous. So you have exactly the opposite situation in those two types of companies.

The contrast that I am giving you illustrates to my mind not only the obvious abuses of the securities market in the last two years, but it also illustrates the fact that the security analyst can in many cases come to pretty definite conclusions that one security is relatively unattractive and other securities are attractive. I think the same situation exists in today's market as has existed in security markets always, namely, that there are great and demonstrable discrepancies in value -- not in the majority of cases, but in enough cases to make this work interesting for the security analyst.

When I mentioned Curtiss-Wright selling at two thirds or less of its working capital alone, my mind goes back again to the last war; and I think this might be a good point more or less to close on, because it gives you an idea of the continuity of the security markets.

During the last war, when you were just beginning with airplanes, the Wright Aeronautical Company was the chief factor in that business, and it did pretty well in its small way, earning quite a bit of money. In 1922 nobody seemed to have any confidence in the future of the Wright Aeronautical Company. Some of you will remember our reference to it in Security Analysis. That stock sold then at eight dollars a share, when its working capital was about \$18 a share at the time. Presumably "the market" felt that its prospects were very unattractive. That stock subsequently, as you may know, advanced to \$280 a share.

Now it is interesting to see Curtiss-Wright again, after World War II, being regarded as presumably a completely unattractive company. For it is selling again at only a small percentage of its asset value, in spite of the fact that it has earned a great deal of money. I am not predicting that Curtiss-Wright will advance in the next ten years the way Wright Aeronautical did after 1922. The odds are very much against it. Because, if I remember my figures, Wright Aeronautical had only about 250,000 shares in 1922 and Curtiss-Wright has about 7,250,000 shares, which is a matter of great importance. But it is interesting to see how unpopular companies can become, merely because their immediate prospects are clouded in the speculative mind.

I want to say one other thing about the Curtiss-Wright picture, which leads us over into the field of techniques of analysis, about which I intend to speak at the next session. When you study the earnings of Curtiss-Wright in the last ten years, you will find that the earnings shown year by year are quite good; but the true earnings have been substantially higher still, because of the fact that large reserves were charged off against these earnings which have finally appeared in the form of current assets in the balance sheet. That point is one of great importance in the present-day technique of analysis.

In analyzing a company's showing over the war period it is quite important that you should do it by the balance sheet method, or at least use the balance sheet as a check. That is to say, subtract the balance sheet value shown at the beginning from that at the end of the period, and add back the dividends. This sum -- adjusted for capital transactions -- will give you the earnings that were actually realized by the company over the period. In the case of Curtiss-Wright we have as much as \$44-million difference between the earnings as shown by the single reports and the earnings as shown by a comparison of surplus and reserves at the beginning and end of the period. These excess or unraveled earnings alone are more than six dollars a share on the stock, which is selling today at only about that figure.